

This assignment will not be graded and is only for practice.

Functions of several variables:

Level 1:

1) Which of the following are correct with respect to $f(x, y) = x^2 - 2y$, where x, y are real? **1 point**

- ☐ Domain of f is $\mathbb{R}^2$
- ☐ Domain of f is $\mathbb{R}$
- ☐ Range of f is $\mathbb{R}$
- ☐ Range of f is $\mathbb{R}^2$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of f is $\mathbb{R}^2$

Range of f is $\mathbb{R}$

2) Let $D \subset \mathbb{R}^n$ and $f : D \rightarrow \mathbb{R}^m, g : D \rightarrow \mathbb{R}^m$ be multivariable functions on D . **1 point**
Which of the following statements is(are) TRUE?

- ☐ The product function fg is defined on D by $fg(\underline{x}) = f(\underline{x}) \times g(\underline{x}), \underline{x} \in D$
- ☐ The sum function $f + g$ is defined on D by $(f + g)(\underline{x}) = f(\underline{x}) + g(\underline{x}), \underline{x} \in D$
- ☐ Let $c \in \mathbb{R}$. The function cf is defined on D by $(cf)(\underline{x}) = \frac{1}{c} \times f(\underline{x}), \underline{x} \in D$
- ☐ If $m = 1$ and $g(\underline{x}) = 0, \underline{x} \in D$, then the function f/g is defined on D by $(f/g)(\underline{x}) = f(\underline{x})/g(\underline{x}), \underline{x} \in D$

No, the answer is incorrect.

Score: 0

Accepted Answers:

The sum function $f + g$ is defined on D by $(f + g)(\underline{x}) = f(\underline{x}) + g(\underline{x}), \underline{x} \in D$

If $m = 1$ and $g(\underline{x}) = 0, \underline{x} \in D$, then the function f/g is defined on D by $(f/g)(\underline{x}) = f(\underline{x})/g(\underline{x}), \underline{x} \in D$

3) Suppose $f : D \rightarrow \mathbb{R}$ is a function defined on a domain $D \subset \mathbb{R}$. Which of the following can be f ?

1 point

☐ $f(x) = 4 \pm \sqrt{x-10}$

☐ $f(x) = \frac{x^2}{x+1}$

☐ $f(x) = \log(\sqrt{x^4+3})$

☐ $f(x) = x^3 + x^2 - x$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x) = \frac{x^2}{x+1}$$

$$f(x) = \log(\sqrt{x^4+3})$$

$$f(x) = x^3 + x^2 - x$$

4) Which of the following statements is(are) TRUE?

1 point

☐ The output of a scalar valued multivariable function is always a real number.

☐ The output of a vector valued multivariable function can be a real number.

☐ The output of a single variable vector valued function is always a vector.

☐ The output of a vector valued multivariable function can not be a real number.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The output of a scalar valued multivariable function is always a real number.

The output of a single variable vector valued function is always a vector.

The output of a vector valued multivariable function can not be a real number.

5) Consider the function $S(t) = (\cos t, \sin t, 2t)$, where t is real. Choose the set of correct options.

1 point

- ☐ S is a single-variable vector-valued function
- ☐ S is a multi-variable vector-valued function
- ☐ The domain of S is $\mathbb{R}$ and the co-domain of S is $\mathbb{R}^3$
- ☐ The domain of S is $\mathbb{R}^3$ and the co-domain of S is $\mathbb{R}^3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

S is a single-variable vector-valued function

The domain of S is $\mathbb{R}$ and the co-domain of S is $\mathbb{R}^3$

6) How many of the following functions is(are) a vector valued multivariable function?

- $f(x, y) = \frac{1}{3\pi}e^{x^2+y^2}$
- $f(x, y) = (x^3, y^2)$
- $f(x, y) = (x + y, x - y, 2xy)$
- $f(x, y, z) = x^2 + xyz - z^2$
- $f(x) = (x + 1, x - 1)$
- $f(x, y, z) = (e^{x+y+z}, 0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

7) Suppose $g : D \longrightarrow \mathbb{R}^3$ is a function defined on a domain $D \subset \mathbb{R}^2$. Which of the following functions may be g ? **1 point**

- ☐ $g(x, y) = 4x + 3y - 12xy$
- ☐ $g(x, y) = (x^4 + 1, y^4 + 1, xy)$
- ☐ $g(x, y, z) = (e^x - e^z, e^z - e^y)$
- ☐ $g(x, y) = (\sin(x + y), \cos(x - y))$

☐ $g(x, y) = (0, \sin(e^x), \cos(e^x))$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$g(x, y) = (x^4 + 1, y^4 + 1, xy)$$

$$g(x, y) = (0, \sin(e^x), \cos(e^x))$$

Level 2:

8) Which of the following are correct with respect to $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$, where x, y are real? **1 point**

- ☐ Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$
- ☐ Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \geq 1\}$
- ☐ Range of f is $[0, 1]$
- ☐ Range of f is $[0, 1)$
- ☐ Range of f is $[1, \infty)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$
Range of f is $[0, 1]$

9) A function $F : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ has the form $F(x, y) = (P(x, y), Q(x, y), R(x, y))$. Choose the correct option(s). **1 point**

- ☐ F is a multi-variable vector-valued function
- ☐ F is a single-variable vector-valued function
- ☐ P, Q, R are single-variable vector-valued functions
- ☐ P, Q, R are multi-variable vector-valued functions
- ☐ P, Q, R are multi-variable scalar-valued functions

No, the answer is incorrect.

Score: 0

Accepted Answers:

F is a multi-variable vector-valued function
 P, Q, R are multi-variable scalar-valued functions

10) Consider $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$. Note: An ellipse is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where a, b are the dimensions of the ellipse. **1 point**

- ☐ Ellipses of increasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1
- ☐ Ellipses of decreasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1
- ☐ Ellipses of constant size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1
- ☐ The curves $\{(x, y) \mid f(x, y) = c\}$, where $c \in [0, 1)$, cannot be represented by an ellipse
- ☐ $f(x, y) = 1$ is a point

No, the answer is incorrect.

Score: 0

Accepted Answers:

Ellipses of decreasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

$f(x, y) = 1$ is a point

11) Consider a point source $S = (1, 2, 3)$ radiating energy. The intensity I at a given point $P = (x, y, z)$ in space is inversely related to the square of the distance d between S and P : $I(x, y, z) = \frac{k}{d^2}$, where k is a real positive constant. Choose the correct option(s). **1 point**

- ☐ Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1\}$
- ☐ Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 2\}$
- ☐ Intensity I is constant on $\{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$
- ☐ Intensity I is constant on $\{(x, y, z) \mid x^2 + y^2 + z^2 = 2\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 1\}$

Intensity I is constant on $\{(x, y, z) \mid (x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 2\}$

Your score is: 0/11